In Search of a Lost Metric: Human Empowerment as a Pillar of Socially Conscious Navigation

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***Abstract*—In social robot navigation, traditional metrics like proxemics and behavior naturalness emphasize human comfort and adherence to social norms but often fail to capture an agent's autonomy and adaptability in dynamic environments. This paper introduces human empowerment, an information-theoretic concept that measures a human's ability to influence their future states and observe those changes, as a complementary metric for evaluating social compliance. This metric reveals how robot navigation policies can indirectly impact human empowerment. We present a framework that integrates human empowerment into the evaluation of social performance in navigation tasks. Through numerical simulations, we demonstrate that human empowerment as a metric not only aligns with intuitive social behavior, but also shows statistically significant differences across various robot navigation policies. These results provide a deeper understanding of how different policies affect social compliance, highlighting the potential of human empowerment as a complementary metric for future research in social navigation.**

***Index Terms*—Human Empowerment; Social Navigation; Social Compliance**

I. Introduction

As robots become more integrated into society, expanding beyond warehouse settings where they handled routine tasks, the need for socially compliant behavior in human environments becomes crucial not only for operational success but also for fostering social and technological sustainability. In an inclusive community where everyone can thrive, robots must navigate their surroundings with greater awareness of social conformity to the humans, ensuring that their behavior respects all individuals and reduces barriers to accessibility. Crowd navigation is one such domain in which research innovations for robots operating in society must be critically evaluated for their impact on society and the environment [1]–[3].

Initially, humans were treated as moving obstacles, but this approach led to issues such as the freezing robot problem and oscillatory behaviors, where robots and humans get stuck in a back-and-forth motion [4]. To address these, more sophisticated methods have been developed, allowing robots to predict human motion in a predict-then-act pipeline [5] or in a coupled fashion with the robot's control action, enabling counterfactual reasoning [6]. Despite these advances, the performance metrics used to assess social compliance have remained largely unchanged, with current research focusing mostly on factors like the robot's arrival time at a goal or whether it can reach its destination without colliding with humans.

Social conformity has primarily been evaluated through concepts such as proxemics, which model personal space around humans and assess whether robots violate these spaces [7], [8]. Some other works have examined how these proxemic boundaries shift as humans move. For instance, rather than modeling personal space as a static ellipse or circle, the space might resemble a cone shape for a moving pedestrian [9]. Other works have focused on encoding specific social navigation rules that humans follow into robot behavior. These include rules for passing, crossing, or overtaking pedestrians [10], as well as guidelines for how robots should interact with groups of people or queues of humans [11]. Readers can refer to [12] for a comprehensive survey of existing metrics.

However, even with these refinements, existing metrics fail to fully measure whether robot movement is socially compliant, as they either try to encode certain behaviors or address this problem in some specific scenarios. To address this gap, we propose a new metric based on the concept of empowerment, which has been mathematically defined in the context of robotics and reinforcement learning. Empowerment relates to an agent's ability to influence its environment in a way that it can observe. In this work, we use the information-theoretic concept of Human Empowerment to assess social compliance in crowd navigation, offering a novel way to evaluate robot behavior in human environments. Using human empowerment, we measure how robot behavior indirectly impacts human sense of agency, specifically assessing how their ability to influence the environment changes. In our setup, human ability is restricted to their motion, with observations represented through an occupancy grid map relative to them.

The social behavior of robots is a broad and interdisciplinary

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topic, covering aspects such as the robot's appearance, user interfaces, and more. In this work, however, we focus exclusively on the social navigation problem and robot mobility, particularly how they navigate within human environments.

In this paper, we introduce a novel approach to evaluating social compliance in crowd navigation by leveraging the concept of human empowerment, an underexplored metric in this context. Our key contributions are as follows:

- **Human Empowerment as a Complementary Metric:** While human empowerment has been mathematically defined in prior work, we propose its potential use as a complementary metric to assess social compliance in crowd navigation. This approach addresses limitations in existing methods that primarily rely on proxemics, which have been stated to be insufficient for comprehensively capturing social behavior in dynamic environments [13].
- **Framework for Integrating Human Empowerment:** We provide a framework that integrates human empowerment into the evaluation of social performance of navigation policies. This serves as a starting point for further research, particularly for human factors researchers interested in the intersection of autonomy and social compliance in robot navigation.
- **Statistical Validation of Human Empowerment:** Through numerical simulations, we show that human empowerment offers insights that align intuitively with social behavior but also reveals statistically significant differences across various robot navigation policies. This highlights the robustness of empowerment as a metric for capturing the nuanced effects of different policies on human autonomy.

By employing empowerment as a metric of performance for social navigation, our work bridges the gap between information-theoretic approaches and the need for socially compliant robot behavior, thus advancing the field of human-robot interaction in dynamic and shared environments.

The rest of the paper is structured as follows: Section II reviews related work. Section III outlines the problem formulation, and Section IV presents the methodology including key aspects of human empowerment. Section V presents and analyzes the evaluation results, while Section VI describes a statistical analysis of different navigation policies on the proposed metric. Finally, Section VII discusses the key findings and concludes the paper.

II. Related Work

A. Social Compliance Metrics in Crowd Navigation

In crowd navigation for robots, several metrics have been developed to assess social conformity, focusing on how well robots adhere to social norms and interact safely with human pedestrians. One such fundamental metric involves proxemics, which evaluates the robot's respect for personal space by measuring distances maintained between the robot and humans during navigation [8]. Another key metric is compliance with the Social Force Model (SFM), which compares the robot's movements to the predicted forces that humans experience, ensuring minimal disruption in pedestrian dynamics [14].

Additionally, pedestrian flow disruption metrics evaluate the extent to which the robot impacts the natural movement of pedestrians, with an emphasis on minimizing trajectory deviations and speed reductions caused by the robot [15]. Human comfort and acceptance are also critical metrics, often assessed through user studies and subjective feedback on how comfortable and safe pedestrians feel around the robot [16]. Goal alignment and social navigation metrics assess how well the robot follows social norms, such as walking on the correct side of a hallway [17], while trajectory predictability measures how naturally human observers can anticipate the robot's future movements [18]. Finally, interaction time is used to evaluate the efficiency of robot-human encounters, with shorter times indicating smoother, socially compliant behavior [19].

However, it remains unclear what navigation behavior represents social optimality [12], and it calls into question whether recreating an exact copy of human motion datasets for behavior naturalness is the ultimate goal. This leads us to an alternate approach of evaluating planning methods developed for social navigation - empowerment. It prioritizes an agent's autonomy by measuring its ability to influence its own environment and maintain control over its future actions.

B. Empowerment

Early research connecting information theory to human-robot interaction (HRI) through the concept of empowerment provided foundational insights into the control and adaptability of autonomous systems. One of the earliest works by Klyubin, Polani, and Nehaniv [20] introduced empowerment as a universal, agent-centric measure of control derived from information theory. Formally, empowerment is defined as the maximum potential for an agent to influence its environment through its actions, quantifiable via Shannon's mutual information between the agent's actions and its future states. This concept laid the groundwork for how robots could use empowerment as a guiding principle to autonomously interact with their surroundings while maximizing their ability to influence future states, which is crucial for adaptive human-robot collaboration.

Building on this, Salge, Polani, and others explored empowerment as an intrinsic motivation for robots in dynamic environments, particularly focusing on sensorimotor systems. In their 2012 study, Salge et al. [21] demonstrated that maximizing empowerment enables robots to "keep their options open," allowing greater flexibility and responsiveness in HRI contexts. This was further elaborated in Salge and Polani's 2013 survey [22], where empowerment was applied as an intrinsic motivation in various empirical studies, showing how robots can maintain meaningful control over interactions with humans by prioritizing flexibility and autonomy. These early works provided a solid theoretical and empirical foundation for using empowerment to specify key robotics principles [23].

Despite rapid developments in the fields of robot navigation and human-robot interactions, the application of empowerment in social navigation remains under-explored [24]. The use

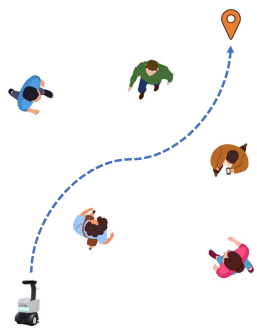

Fig. 1: A robot navigating in a crowded environment.

of empowerment as an agent-centric measure could help robots balance autonomy and social conformity by optimizing their ability to navigate dynamic human environments without sacrificing control over their own future actions.

III. Problem Formulation

In this study, we investigate a navigation scenario within a 2-D environment $\mathcal{W} \subset \mathbb{R}^2$, where a single robot operates among $N \in \mathbb{R}$ human pedestrians, as illustrated in Fig. 1. We let $\mathcal{H} = \{1, \ldots, N\}$ represent the set of humans in the scene. We define the position and velocity of any agent, robot or human, using Cartesian coordinates as $\mathbf{s}_k = [z_k^x, s_k^y]^\top \in \mathcal{W}$ and $\mathbf{v}_k = [v_k^x, v_k^y]^\top \in \mathbb{R}^2$, respectively, with $k \in \mathbb{N}$ representing the discrete time-step and x, y the axis components. The state of the robot and its control action at time-step k are denoted by $\mathbf{x}_k = [\mathbf{s}_k^\top, \mathbf{v}_k^\top]$ and $\mathbf{u}_k = \mathbf{v}_k$, respectively. A superscript R or H is added below to indicate whether the state of interest refers to that of a robot or a human respectively.

The general objective of social robot navigation is to efficiently navigate the robot from its initial position to its goal location within a time limit. During navigation, the robot must avoid colliding with other humans while minimizing disruptions to their intended trajectories, which can cause discomfort. This paper focuses exclusively on motion planning, assuming the robot is equipped with onboard sensors that provide the real-time positions of surrounding human agents in the vicinity.

IV. Methodology

A. Mutual Information

Mutual Information (MI) is a key concept in information theory that quantifies the amount of information shared between two random variables. It measures how much knowing the value of one variable reduces the uncertainty about the other, making it a versatile metric for assessing both linear and nonlinear dependencies.

For two random variables X and Y, mutual information is defined as:

$$I(X;Y) = \int \int p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right) dx\, dy \tag{1}$$

where $p(x, y)$ is the joint probability distribution of X and Y, and $p(x)$ and $p(y)$ are the marginal probability distributions of X and Y, respectively. This formulation shows that MI is derived from the Kullback-Leibler divergence between the joint distribution $p(x, y)$ and the product of the marginal distributions $p(x)p(y)$. In other words, MI measures the difference between the actual joint distribution and what the distribution would be if the variables were independent.

In planning, MI can be computed between components such as states, actions, and future states, depending on the behavior analyzed. For example, $I(s_t; a_t)$ measures the information the current state s_t provides about the action a_t. Similarly, $I(a_t; s_{t+1})$ captures the influence of the action a_t on the resulting future state s_{t+1}, while $I(s_t, a_t; s_{t+1})$ evaluates the combined influence of the state-action pair (s_t, a_t) on the future state s_{t+1}. This provides insights into the agent's decision-making process and control over the environment, making MI a powerful metric for evaluating how decisions and states impact future outcomes, valuable for analyzing control strategies.

B. Human Empowerment

Empowerment is a concept rooted in information theory that measures the extent to which an agent can influence its environment. It is defined as the mutual information between the agent's actions and the resulting future states of the environment. Essentially, it quantifies the agent's control over its surroundings. Unlike traditional optimization metrics, which often focus on immediate or specific objectives, empowerment emphasizes maintaining the agent's potential to influence future states, promoting behaviors that preserve adaptability and open possibilities for future decisions.

Formally, empowerment for an agent can be expressed as $E(s) = \max_\pi I(a_t; s_{t+1}|s_t)$, where a_t represents the actions the agent can take at time t, s_t is the current state, and s_{t+1} is the resulting future state and π is the agent's policy. Optimization over policies ensures that the agent selects actions that maximize its influence over future states. This approach not only enables the agent to achieve its current objectives but also retains flexibility to adapt and make meaningful decisions as the environment evolves.

Following this, **Human Empowerment** extends the concept of empowerment to human-robot interactions, focusing on quantifying how much a human's future state is influenced by the actions they take. In this context, human empowerment measures the degree to which a human's actions affect their future states, serving as a metric for assessing how much empowerment is gained throughout a trajectory. Rather than being tied to any specific goal or objective, this metric focuses purely on the extent of influence exerted by a human's actions.

Human empowerment can be expressed as the mutual information between the human's actions and the resulting future states in the environment:

$$E(z_t^H) = \max_w I(a_t; z_{t+1}^H | z_t^H), \tag{2}$$